

View results

Respondent

49

Anonymous

154:12

Time to complete

A. General information about the respondent

1. * Country

Cuba



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Science, Technology and Environment, CITMA

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://www.citma.gob.cu>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The Ministry of Science, Technology and Environment is the body that has the mission of proposing and, once approved, directing and controlling the State and Government policy in the areas of science, technology, innovation, environment and others that are determined to contribute to the sustainable development of the country.

7. * Full name of officially designated contact person(s) who completed this survey

Ricardo Casate Fernández.

8. * Position of officially designated contact person(s) who completed this survey

Director of Strategic Programs and Projects

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

In preparing this national report on the implementation of the 2021 Open Science Recommendation, the roadmap described in the distributed Operational Guide was followed. A work team was formed led by the General Directorate of Science, Technology and Innovation of CITMA and made up of representatives of organizations, research institutions, professional associations; as well as the scientific community and experts. A preparatory workshop for the process was convened, agreeing to apply a survey made up of the elements of the questions from the questionnaire provided. Subsequently, the focus groups met to delve deeper into the results of the application, to detail and shape the draft report. In a subsequent session, the information reported in this First National Report on the Implementation of the 2021 Recommendation on Open Science was reviewed and defined.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Ministry of Higher Education (MES), Ministry of Education (MINED), Central University of Las Villas "Marta Abreu", University of Pinar del Río, University of Camagüey "Ignacio Agramonte Loynaz" (UC), University of Havana, Infomed - National Center for Medical Sciences Information, Institute of Tropical Geography, Cuban Society of Philosophical Sciences, Cuban Society of Botanic (Socubot), Cuban Society for the Promotion of Renewable Energy Sources and Environmental Respect (Cubasolar), Cuban Society of Chemistry, Cuban Society of Information Sciences, Nuclear Energy and Advanced Technologies Agency (AENTA), CERVANTES-DIGITAL PRODUCTIONS SRL.

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<http://www.mes.edu.cu>, <https://www.mined.gob.cu/>,
<https://www.uclv.edu.cu/>, <https://www.upr.edu.cu/home>, <http://www.reduc.edu.cu>,
<https://www.uh.cu/>, <https://www.uho.edu.cu/>, <http://www.infomed.sld.cu/>,
<https://www.geotech.cu/>, <https://www.academiaciencias.cu/es/node/465>, <https://www.ipscuba.net/directorio-sc/sociedad-cubana-de-botanica-socubot/>,
<http://www.cubasolar.cu/>, <http://quimred.fq.uh.cu/scq>, <http://www.socict.org/>,
<http://www.aenta.cu>, <http://ojs.edicionescerantes.com>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- ☒ Public
- ☒ Private
- ☒ Civil Society
- ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Science, Technology, and Environment (CITMA), Ministry of Higher Education (MES), Ministry of Public Health (MINSAP), Ministry of Education (MINED)

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Spanish

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

National and international events have been organized, as well as meetings and workshops related to open access for scientific publications, as well as the importance of using platforms for the implementation of institutional repositories, science evaluation systems, integrated systems for library management, open educational resources, free software, and the management of research data: □ First Symposium on Open Science in Cuba (2023) Within the framework of the International Scientific Convention "Saber UH 2023" of the University of Havana, from May 29 to June 2. <https://convencionuh.com/index.html> The organization of its second edition is underway in April 2025. □ Meetings and workshops within the framework of the OLINFER 2023 event of the international project ELINF – RedTlc VLLr. <https://olinfer2023.vlired.cu/> □ Within the framework of the UCLV 2023 International Scientific Convention, <https://convencion.uclv.cu/es/event/ii-international-symposium-on-generation-and-transfer-of-knowledge-for-digital-transformation-sitic2023-i-international-workshop-on-technologies-and-applications-of-semantic-web-in-the-artificial-intelligence-environment-of-things-181/register> . Abstract Report: <https://convencion.uclv.cu/es/memorias-cci-2023> □ First International Workshop on Open Science in Higher Education. Within the framework of the 13th International Congress on Higher Education 'University 2024' from February 7 to 11, 2024. <https://www.congresouniversidad.cu/event/i-taller-internacional-sobre-ciencia-abierta-en-la-educacion-superior-ca-82/register> □ Workshop on scientific journals within the framework of the Animal Production and Agrodevelopment Convention, Agropat 2022, from October 10 to 14, 2022. This event was called by the Institute of Animal Science (ICA) □ Pre-congress Workshop 6th International Agrodevelopment Convention 2022, organized from October 24 to 28, 2022 by the University of Matanzas and the Indio Hatuey Pasture and Forage Experimental Station. <https://www.agrodesarrollocuba.com/> □ National Workshop on Scientific Publication in Health Sciences. Organized by Medical Sciences Publishing House (ECIMED) and sponsored by the National Center for Medical Sciences Information and the National Council of Scientific Health Societies in the years 2022, 2023, and 2024 with topics related to open science. <https://publicient.sld.cu> □ International Symposium "Open Science for Equity". Held from January 28 to 31, 2025, organized by the Cuban Academy of Sciences. This event seeks to promote dialogue on open science, addressing issues such as international collaboration, exchange of best practices and equitable access to scientific knowledge. It is intended to facilitate the creation of networks between researchers and social actors at a global level. <https://www.academiaciencias.cu/es/node/1302>

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☐ Yes

☒ No

23. 1.4. cont. If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Work is being done on initiatives that are still incipient and independently by institutions interested in incorporating values and principles into the conduct of their research activities.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

At the end of 2024, the system policy on science, technology and innovation was updated, which will serve as the basis for the development of the science, technology, and innovation law and includes several components of open science, particularly open scientific knowledge.

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☐ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

The resolution with the greatest implications for open science is the Resolution 149 of July 1, 2024 issued by CITMA establishing the National Certification System for Scientific Journals. The system is defined as the structured set of processes and activities aimed at increasing the scientific and editorial quality, visibility, recognition, and professionalization of Cuban scientific journals. Among them, the process of certification and categorization of scientific journals is implemented through the calls made by the organization annually and is executed by the National Evaluation Commission chaired by its Director of Scientific and Technological Potential. The indicators established for the evaluation correspond to those established by the indexing systems of Redalyc, the Latindex, 2.0 catalog, SciELO and the Directory of Open Access Journals.

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☐ Yes
- ☒ Under development
- ☐ No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

They are under development: ☐ Policy proposal for Research Data Management at UCLV https://www.researchgate.net/publication/380515811_Policy_proposal_for_Research_Data_Management_at_UCLV Open Science Policy for the Ministry of Higher Education. <https://zenodo.org/records/14884420>

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☐ Under development
- ☒ No

32. 2.4 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/_page=9

In Cuban institutions and universities, the implementation of open science is carried out through various programs, methods, and mechanisms that promote transparency, accessibility, and collaboration in scientific research. This is largely due to the impact of the ELINF project on open access within Cuban Higher Education. <http://dspace.uclv.edu.cu/handle/123456789/12241>. For example:

- Creation of digital repositories as part of the MES Institutional Digital Repository Network (Resolution No. 30/2024), Policy for the Institutional Digital Repository Network of the Ministry of Higher Education is attached.
- Massive Open Online Course (MOOC) on Open Education and Science: This course, developed by the ELINF-VLIR project, offers training on topics such as copyright, open educational resources, research data management, and open science evaluation.
- Open Science Policy for the Cuban Sports System: Implemented in 2023, this policy seeks to develop key components of open science, such as open access, research data management, open peer review, and the necessary infrastructure. It also includes funding mechanisms and incentives for its promotion. http://forumcidc.inder.gob.cu/wp-content/uploads/2023/05/POLITICA-CIENCIA-ABIERTA-INDER-GRIZZLY-MENESES-UCLV.pdf?utm_source=perplexity
- Strategy and Policy for Open Science and Education: Presented as a draft in 2019, it seeks to establish a strategic framework that promotes openness in scientific and educational production in the institutions of the Ministry of Higher Education. It includes guidelines for the creation of digital repositories and the implementation of open science practices. https://www.congresouniversidad.cu/event/i-taller-internacional-sobre-ciencia-abierta-en-la-educacion-superior-ca-82/track/recomendacion-de-educacion-y-ciencia-abierta-para-las-universidades-cubanas-5127?utm_source=perplexity

Within the strategic planning of the MES, indicators that contribute to open science have been incorporated, to be measured in all universities. For example:

- Massive open online courses (MOOC) available in Virtual Teaching Learning Environments (EVEA).
- Training actions for teachers in setting up courses in virtual environments.
- Actions carried out for the development of digital skills in students, teachers, researchers and non-teaching staff
- Institutional Digital Repositories registered in OpenDOAR

Scientific associations have also conducted actions related to practices, values and benefits of open science:

- Cuban Society of Information Sciences This platform aims to manage different documentary collections belonging to scientific fields, research associations or scientific societies and make them available to national and international users through the official SOCICT website and through social networks on the Internet. <http://societyopen.society.org/>
- Cuban Society of Law and Informatics In the adoption of the National Policy for Digital Transformation and the Cuban Digital Agenda, as well as the Strategy for the development and use of Artificial Intelligence. It addresses areas of interest for the adoption of open science, including: inclusive and participatory digital society, secure and quality connectivity, responsible and secure use of cyberspace, reference frameworks and architectures (conceptual, organizational, data, semantic, etc.) and governance model, the implementation of disruptive technologies such as the Internet of Things, Artificial Intelligence, Big Data and others.

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☐ Yes
- ☐ Under development
- ☒ No

36. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

It is not institutionalized as a mechanism for these purposes. However, funding has been provided for research projects explicitly dedicated to the implementation of open science or some of its components: Sectorial program "Management of science, technology, and innovation for sustainable development" managed by the Agency for Social and Humanistic Sciences, which finances the following projects: <https://instituciones.sld.cu/inhem/files/2023/03/CONVOCATORIA-PSCTI-ACSH.pdf> □ Project Open Science Ecosystem in the Institutions of the Ministry of Higher Education (2024-2026) head by University Central de las Villas "Marta Abreu" <https://www.facebook.com/ECAMESCUba> □ Project Research Data Management in Cuban Higher Education and Scientific Institutions Educational Institutions head by Central University of Las Villas "Marta Abreu" <https://www.facebook.com/groups/832040065642742> □ Project Strategy for the implementation of open education in the Ministry of Higher Education of Cuba head by Central University of Las Villas "Marta Abreu" (Starts 2025-) □ Project "Digital transformation of the management and evaluation of science, technology and innovation activities in Cuba Science in Cuba" head by institute of Scientific and technological information (IDICT) □ Project "Data Management Model of the Scientific, Technological and Innovation Observatory" head by institute of Scientific and technological information (IDICT)

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☐ Yes

☒ No

38. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

At the end of 2024, the system policy on science, technology and innovation was updated. It is not institutionalized as a mechanism for these purposes. However, funding has been provided for research projects explicitly dedicated to the implementation of open science or some of its components

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

CERVANTES-DIGITAL PRODUCTIONS SRL. <http://ojs.edicionescervantes.com> SMEs created in accordance with current Cuban legislation Certified in 2019 by SCielo-México as an editor specialized in the production of XML-JATS in accordance with the Publishing Schema methodology.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☐ Yes
- ☐ Under development
- ☒ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

42. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The most recent data on the percentage of national gross domestic product (GDP) that our country spends on research and development expenditures available is 0.32 in the year 2021. <https://databrowser.uis.unesco.org/view#atorPaths=&geoMode=countries&geoUnits=&timeMode=range&view=table&chartMode=multiple&chartHighlightSeries=&chartHighlightEnabled=true&indicatorPaths=UIS-SDG9Monitoring%3A0%3AEXPGDP.TOT>

43. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

The proportion of the population with Internet access according to the data available for the year 2024, it is 5.35 M. <https://datahub.itu.int/data/?e=CUB&c=701&i=11632&v=chart>

44. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

45. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

National HPC Network Making Supercomputing Available to All Cuban Researchers. Computing in Science & Engineering. May/June 2018. IEEE CS and AIP. 1521-9615/18. In close collaboration between the Universidad Central de las Villas "Marta Abreu" and the Belgian University of Ghent. Cubaeduca <https://www.cubaeduca.cu/about> An online educational community managed by CINESOFT, serving Cuban and international education. As an integrate and interactive platform, it offers a set of content, resources and services related to education. Cuban Science Network. <http://redciencia.cu/> This digital platform includes a wide range of information resources and services on science, technology, and innovation available in the country, so that scientists, technologists, teachers, and the general public can access and benefit from this information. Infomed <https://www.sld.cu/about/> Infomed is the name that identifies the network of people and institutions that work and collaborate to facilitate access to information and knowledge, necessary to improve the health of Cubans and the peoples of the world. Biological Corridor in the Caribbean <https://cbcbio.org/> The Integrated Knowledge Management System (IKMS) is a web platform integrated into the information system of the Caribbean Biological Corridor, CBC. protection of biodiversity. It includes: BioAtlas – Geographic Information Tool of the CBC Initiative, CBC Databases – Management tool for the CBC Executive Secretariat, EVEA-CBC – Tool for Virtual Courses, VAC-CBC – Surveillance and Intelligence Service Latin American Network for Education in Nuclear Technology (LANENT) <https://www.iaea.org/es/servicios/red-> The Latin American Network for Education in Nuclear Technology was established to contribute to preserving, promoting and sharing nuclear knowledge as well as fostering nuclear knowledge transfer in the Latin American region. The Latin America and the Caribbean Network (RALACA) <https://www.iaea.org/es/servicios/red-analitica-de-latinoamerica-y-el-caribe-ralaca> Brings together analytical laboratories in Latin America and the Caribbean to enhance regional capabilities to target food safety and environmental sustainability. REMARCO "Research Network of Marine-Coastal Stressors in Latin America and the Caribbean" <https://remarco.org/remarco/> Is a voluntary, non-profit, interdisciplinary research network that uses nuclear and isotopic techniques to address environmental problems of the marine-coastal ecosystems of Latin America and the Caribbean. REPROLAM. Latin American Network for the Optimization of Occupational Radiation Protection. <http://www.reprolam.com/> It is defined as a regional ALARA network on the ORPNET platform, the IAEA network that aims to promote the optimization of occupational radiation protection. Its operation is supported by a dynamic environment of development of information technologies and social networks. IAEA/WHO Network of Secondary Dosimetry Calibration Laboratories (SSDL Network) The joint IAEA/WHO SSDL Network was established in 1976 to improve the accuracy of radiation dosimetry. The Latin American Biological Dosimetry Network (LBDNet) Was constituted in 2007 for mutual assistance in case of a radiation emergency The main objectives are: to strengthen the technical capacities of Biological Dosimetry Services belonging to laboratories existing in the region.

46. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

47. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:



federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage



community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society



platforms for exchanges and co-creation of knowledge between scientists and society



community-based monitoring and information systems to complement national, regional and global data and information systems

48. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

All are non-commercial infrastructures and compliance with the Recommendation, FAIR and CARE principles

SCIENTIFIC MAGAZINES 124 journals are registered in the Open Access Journal Directory [https://doaj.org/search/journals?](https://doaj.org/search/journals?source=%7B%22query%22%3A%7B%22query_string%22%3A%7B%22query%22%3A%22Cuba%22%2C%22default_operator%22%3A%22AND%22%2C%22default_fiel)
[source=%7B%22query%22%3A%7B%22query_string%22%3A%7B%22query%22%3A%22Cuba%22%2C%22default_operator%22%3A%22AND%22%2C%22default_fiel](https://doaj.org/search/journals?source=%7B%22query%22%3A%7B%22query_string%22%3A%7B%22query%22%3A%22Cuba%22%2C%22default_operator%22%3A%22AND%22%2C%22default_fiel)
[d%22%3A%22index.country%22%7D%7D%2C%22size%22%3A50%2C%22sort%22%3A%5B%7B%22created_date%22%3A%7B%22order%22%3A%22desc%22%7D%7D%5D%2C%22track_total_hits%22%3Atrue%7D](https://doaj.org/search/journals?source=%7B%22query%22%3A%7B%22query_string%22%3A%7B%22query%22%3A%22Cuba%22%2C%22default_operator%22%3A%22AND%22%2C%22default_fiel) 53 journals are registered in Redalyc, of them 25 journals in AmeliCA <https://www.redalyc.org/pais.oa?id=35> 489 journals are registered in LATINDEX, of them 78 journals in the 2.0 Catalog https://www.latindex.org/latindex/bAvanzada/resultado?idMod=0&titulo=&otrostitulos=&issn=&tema=0&subtema=0&editorial=&idioma=0&alicio=&aFin=®ion=0&pais=12&ciudad=&estado=&natPub=0&natOrg=0&situacion=0&frecuencia=0&soporte=&arbitrada=&acc_abierto=&derechos=&cobro=&acceso=&cobertura=0&f_unico=&send= Buscar 232 certified journals by CITMA of them 50 in Redalyc and 20 in AmeliCA

DIGITAL REPOSITORIES OpenDOAR. 18 records

https://v2.sherpa.ac.uk/view/repository_by_country/Cuba.html 1. AlmaA. <https://rc.upr.edu.cu/> 2. <http://dspace.uclv.edu.cu/> 3. EHTC Institutional Repository. <http://www.repositorio.ehtc.cu/jspui/> 4. Institutional Repository of the Ministry of Higher Education (Cuba) <http://repositorio.eduniv.cu> 5. Nineve. <http://ninive.ismm.edu.cu/> 6. ReDIUC. <http://rediuc.reduc.edu.cu/> 7. Institutional Digital Repository of the University of Holguín. <https://repositorio.uho.edu.cu> 8. Institutional Repository of the University of Matanzas. <https://rein.umcc.cu> 9. Cuban Environmental Information Repository. <http://repositorio.geotech.cu/jspui/> 10. Repository of Health Legislation in Cuba. <http://legislacion.sld.cu/> 11. Thesis Repository. <https://repositorio.cujae.edu.cu/home> 12. Repository of PhD Theses in Biomedical and Health Sciences of Cuba. <http://tesis.repo.sld.cu/> 13. Repository of Scientific Publications. <http://www.repoarticulos.art.sld.cu/> 14. Thesis Repository. <http://www.repotesis.art.sld.cu/> 15. Institutional Repository of the University of Guantanamo. <http://repositorio.cug.co.cu:8080/jspui/> 16. Scientific Electronic Library Online - Cuba. <http://www.scielo.sld.cu/> 17. Scriptorium. <https://accesoabierto.uh.cu/s/scriptorium> 18. UNISS Institutional Repository. <https://dspace.uniss.edu.cu/> ROAR* 12 records <https://roar.eprints.org/view/geoname/geoname=5F2=5FCU.html> 1. CREATIIF Digital Photo Library of the University of Havana. <https://accesoabierto.uh.cu/s/creatiif> 2. SciELO - Cuba. <http://www.scielo.sld.cu/scielo.php> 3. EcuRed Virtual Library. <http://revistas.mes.edu.cu/greenstone/cgi-bin/library.cgi> 4. Cuban Environmental Information Repository. <http://repositorio.geotech.cu/jspui/> 5. AlmaA. <https://rc.upr.edu.cu/> 6. <http://dspace.uclv.edu.cu/> 7. EHTC Institutional Repository. <http://www.repositorio.ehtc.cu/jspui/> 8. UNISS Institutional Repository. <https://dspace.uniss.edu.cu/> 9. ReDIUC. <http://rediuc.reduc.edu.cu/> 10. Nineveh. <http://ninive.ismm.edu.cu/> 11. Institutional Digital Repository of the University of Holguín. <https://repositorio.uho.edu.cu/> Open Education: <https://aula.uclv.edu.cu/>. Scientific production in Open Access: <https://dspace.uclv.edu.cu/home>. Information System for Science (in development): <https://vivo.uclv.edu.cu/>. The Sceiba Project, developed by the University of

Havana. [https:// sceiba.reduniv.edu.cu/](https://sceiba.reduniv.edu.cu/) The University of Havana magazine portal.
<https://revistas.uh.cu/>

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

50. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☒ Yes

☐ No

☐ Other

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

52. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

REDALYC. <https://www.redalyc.org/> The Network of Scientific Journals of Latin America and the Caribbean, Spain and Portugal (Redalyc) is an indexing system that integrates into its index the journals of high scientific and editorial quality in the region. After 16 years of giving visibility and supporting the consolidation of journals, it now exclusively integrates those that share the non-profit publication model to preserve the academic and open nature of scientific communication, from any region. LATINDEX <https://www.latindex.org/latindex/> Network of 24 institutions that work in a coordinated manner to gather and disseminate information on scientific journals produced in the region, also integrating those that disseminate Ibero-American studies throughout the world. Registration and consultation are free. There are 489 journals indexed in the Directory, 78 of which are in the 2.0 catalogue. SCIELO <http://www.scielo.sld.cu/scielo.php> Cuba-AmeliCA National Collaboration Program: This initiative seeks to strengthen open science in Cuba through a cooperation agreement with AmeliCA, which promotes open access and scientific visibility. The program includes actions such as support for scientific journals and the creation of a National Repository of Scientific Journals. CLACSO https://www.clacso.org/en/institucional/centros-asociados/?pag=pais&id_country=12&ct=0 The Latin American Council of Social Sciences (CLACSO) is an international non-governmental institution with associative status at UNESCO. Its objectives include generating open access conditions to Latin American and Caribbean academic production, contributing to the democratization of access to knowledge and allowing its more active use by public policy makers, social and citizen organizations, the press and the university system itself. There are 41 Cuban institutions associated with the organization, 34 with full member center status, and 7 associated organizations. Ameli-Open Knowledge (AmeliCA) <http://amelica.org/>. Initiative founded by UNESCO, Redalyc and CLACSO for academic publication non-commercial, AmeliCA promotes open science in an open and collaborative manner, both in relation to research processes and work instruments as well as intermediate and final results, through methodologies, tools and infrastructures that facilitate the opening of processes and allow the reuse, distribution and reproduction of research.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

The process has been carried out with several initiatives and projects: □ Subjects in the undergraduate course on Research Data Management
<https://moodle.uclv.edu.cu/course/view.php?id=12185> □ Course on Research Data Management in the Master's Degree in Information and Knowledge for Development
<https://aula.uclv.edu.cu/course/view.php?id=488> □ Data Management Training.
<https://olinfer2023.vlired.cu/?p=1988> □ MOOC on Open Education and Science:
<https://moodle.vlired.cu/course/view.php?id=56>
<https://dspace.uclv.edu.cu/server/api/core/bitstreams/1d89ecb2-802c-42cf-8277-561fb224ccdd/content> □ Elective subjects taught in the Information Sciences program at the Faculty of Communication of the University of Havana: Open Access (2 editions) and Open Science (1 editions) □ Draft Strategy and Policy for Open Science and Education: This document, prepared by the Ministry of Higher Education (MES), seeks to establish a regulatory framework to promote open science in Cuban universities. Although a formal national policy has not yet been implemented, this draft is a significant step towards structuring capacities in this area. □ Projects such as ELINF and Sceiba: These projects have been fundamental in promoting open science in Cuba. ELINF focuses on the training and development of human resources, while Sceiba seeks to promote open access and improve scientific infrastructure. □ The project "Information and Communication Technologies (ICTs) at the service of educational processes and knowledge management in Higher Education" (ELINF) of the University of Camagüey, part of the Program "Strengthening the role of ICTs in Cuban universities for the development of society" (REDTIC), in which universities have benefited from Cuban-Flemish collaboration, offers opportunities, especially for improving the quality of teaching and research and thus demonstrating their transversality in Higher Education. This project has included the implementation of massive open online courses (MOOCs) on open education and science, as well as the creation of research information systems to facilitate scientific visibility and collaboration. □ Workshops, seminars and training have been held for researchers and academics to promote understanding of the principles of open science and their implementation in practice. Some institutions have also adopted internal policies that encourage open access to research publications and data.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☒ Yes

☐ No

58. 4.3 cont. **If yes**, please provide more information and links as relevant.

The improvement of the higher education system is under development, but there are already courses incorporated into the programs: Subjects in the undergraduate course on Research Data Management <https://moodle.uclv.edu.cu/course/view.php?id=12185> Elective subjects taught in the Information Sciences program at the Faculty of Communication of the University of Havana Open Access (2 editions) Open Science (1 editions)

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

61. 4.5 cont. **If yes**, please provide more information and links as relevant.

Scientific Observatory <http://www.youtube.com/c/ObservatorioCient%C3%ADfico>
[Facebook.com/ObservatorioCientifico](https://www.facebook.com/ObservatorioCientifico) twitter.com/ObserveCiencia The Scientific Observatory, a 27-minute program on Canal Caribe, is part of the "Transmedia Project for the Communication of Science, Technology, Innovation and the Environment in Cuba," sponsored by the Ministry of Science, Technology and Environment (CITMA) and the Financial Fund for Science and Innovation (FONCI). The purpose of the Observatory is spreading diverse scientific issues and it has a range of its own content on digital platforms. Cultivar Conciencia
<http://www.youtube.com/@cultivarconciencia> The Cultivar Conciencia program emerges as an audiovisual project that aims to connect science with agriculture, which is on television and social networks every Saturday to share various topics, considering the current challenges for those who research and those who produce the land in Cuba. Juventud Técnica.
<https://www.juventudtecnica.cu/JuventudTecnica.cu> It is a Cuban publication dedicated to the dissemination of science, technology, innovation and environmental issues. Currently affiliated with the Casa Editora Abril, it is a multi-platform medium, with profiles on the main social networks that offers a variety of digital products and services. It seeks to disseminate scientific-technological and environmental issues, and serve as a space for exchange on these. In addition, it informs and reflects on the work of young Cuban scientists and technologists and the general development of science and innovation in Cuba, Latin America and the rest of the world. Scientific Communication Award: The University of Holguín established a Scientific Communication Award to recognize efforts to disseminate research results and promote university science. This initiative seeks to raise scientific culture among the population and make scientific achievements visible. National Certification System for Scientific Journals: This system aims to raise the quality and international visibility of Cuban scientific journals, contributing to a better dissemination of the scientific results generated in the country. The Tocaroro Project of the University of Pinar del Río offers a virtual space for the comprehensive management of Cuban scientific production, contributing to improving its access and visibility. It facilitates the interrelation between all the actors involved in the processes related to scientific research and also encourages a broader dialogue between the scientific community and society, thus strengthening the culture of shared knowledge. The creation of institutional and national repositories where scientific publications and research data can be deposited and shared across all universities. The Scriba Project, developed by the University of Havana, aims to identify what has been done and what initiatives universities can promote open science and how to contribute to generalizing its practice in the country. It includes the organization of symposiums and the publication of summaries and proceedings of conventions. It has become a platform developed for quality control and monitoring of Cuban scientific publications, promoting open science. Cuban Academy of Sciences and various scientific associations have developed projects aimed at improving scientific communication, fostering interdisciplinary dialogues and promoting the dissemination of results in various areas of knowledge. For example, Open Doors, organized every month to listen to and exchange ideas about science, the environment and technology, brings together scientists, environmentalists, students, teachers and interested people <https://www.citma.gob.cu/destacan-rol-de-puertas-abiertas-de-academia-de-ciencias/>

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

SciELO Cuba which is the electronic virtual library of Cuban scientific journals on medicine as part of the SciELO Network of Ibero-America. The main transformations and proposals currently being worked on are declared: the encouragement of the use of preprint servers, called to accelerate the communication of science and its recognition; the use of research data repositories to promote their openness and ensure their preservation, reusability, evaluation and reproducibility; the use of unique identifiers for the control of authors (ORCID, www.orcid.org), and continuous publication as an immediate variant in the delivery of information. http://scielo.sld.cu/scielo.php?pid=S2307-21132019000200001&script=sci_arttext <https://talleres.ucf.edu.cu/event/13/contributions/319/> <https://zenodo.org/records/14884602>

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

65. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

67. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☒ YES, at both the national and institutional levels
- ☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☐ Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

Promotion of preprints We have the case of SciELO Cuba, which declares that the main transformations and proposals currently being worked on include encouraging the use of preprint servers. Alfonso MJE. Open science: a new stage for the Cuban Journal of Health Sciences Information. Cuban Journal of Health Sciences Information (ACIMED). 2019;30(2):1-3. http://scielo.sld.cu/scielo.php?pid=S2307-21132019000200001&script=sci_arttext Promotion of open innovation practices. Innovation Laboratories for the Digital Transformation of Culture of the Union of Computer Scientists of Cuba. <https://co-lab.uic.cu/> Strengthening the role of the UIC in the digital transformation of culture at a national level, creating a network of citizen innovation laboratories and strengthening the capacities of those involved in the 5 selected territories, facilitating citizen participation, developing, and validating prototypes of solutions in the field of local cultural heritage. <https://fundacioninnova.uh.cu/> Innovation and Development Foundation. <https://fundacioninnova.uh.cu/incuba/> Incubator for innovation projects at the University of Havana with the aim of increasing the impact of scientific results on the Cuban economy and society. It has designed spaces to foster the entrepreneurial spirit where success stories and good practices are presented; to disseminate the work carried out among university students to foster a culture of innovation from their training; to socialize with the aim of generating effective interactions between the academy and external actors; and for all types of actors to shape their ideas, solve real challenges, and experiment with solutions.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Government participation in regional, subregional and international initiatives: The Latin American Open Science Forum 2024 from September 11-13. The event served to reinforce the commitment to link science to the needs of people and the planet, and to its constant struggle for access to the benefits of scientific progress for all humanity. Caracas Declaration: regional launch of the UNESCO Recommendation on Open Science from Latin America and the Caribbean. It was agreed to consider knowledge as "a human right and duty and, therefore, a common good of the peoples that must be at the service of the needs of humanity", so they agreed to promote more exchanges in the region. And the establishment of alliances between scientific actors and communities, which allow the generation of appropriable knowledge for the transformation of capacities in the region." <https://unesdoc.unesco.org/ark:/48223/pf000038474> Diamond Global Summits on Open Access. Led by Redalyc in Mexico in 2023 and in Cape Town in 2024. <https://globaldiamantaoa.org/>. <https://doasummit.uct.ac.za/> Cooperation Agreements: Cuba has established bilateral and multilateral agreements with other countries and institutions, promoting the exchange of knowledge, resources and experiences in the field of open science. Participation in International Collaborative Networks: The country and its scientific institutions have joined international networks and platforms that promote open science, allowing Cuban researchers to access information, projects and global collaborations. Joint Projects: Joint research projects have been developed with foreign institutions, where data, methodologies and results are shared, facilitating a collaborative approach to research. National Cuba-AmeliCA Collaboration Program for the creation of the National Repository of Journals Scientific collaboration in health, including bilateral and multilateral projects, such as that between the Center for Molecular Immunology in Cuba and the Roswell Park Institute in Buffalo, New York, for research in biotechnology against cancer. Internationally funded project (VLIR- Belgium): Development of a research data management strategy in Higher Education and Research Institutes in a Latin American context (2022-2024) Internationally funded project (VLIR- Belgium): (Open) education in research data management to accelerate open and responsible science policy and implementation in Higher Education Institutions (2025-2026) International Events: Cuba organizes and participates in international conferences, symposia and workshops that address open science issues, allowing for the exchange of ideas and the creation of links between scientists from different countries. Open Access: The incipient developments of institutional policies encourage publication in open access journals, which makes it easier for the results of Cuban research to be accessible to a global audience.

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

74. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

76. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

78. 8.1 cont. If yes, please provide more information.

United States Government Economic Blockade: This is considered the main obstacle to scientific development in Cuba. A product of this unjust policy: □ Scientific researchers, academics and the Cuban population in general cannot access certain sites and services that facilitate collaboration, exchange and training that include open science topics. In addition, they cannot access specialized platforms for videoconferencing that facilitate exchange, training and collaboration (Zoom). □ Cuban platforms are unable to use the DOI, one of the most widely used persistent identifiers of digital objects in open science infrastructures, which affects their interoperability. They are also unable to use computer programs that, despite being distributed under open licenses, are restricted for download and use in our country. □ Cuba's access to international scientific and academic cooperation and the effective participation of researchers in events, organizations and networks that allow the exchange of information, knowledge and good scientific practices are hindered. □ Access to international funding, technologies, and resources necessary for research is hampered. This not only limits the capabilities of Cuban researchers, but also affects institutions in other countries, especially in the United States. □ The acquisition of scientific technology, both generic and specific, is becoming more difficult, and the obsolescence of the available infrastructure is becoming more acute. □ Access to international databases, publications and specialized sites is limited, affecting the ability of Cuban researchers to stay up to date with global advances in their fields.

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

Despite the mentioned progresses in areas related to the promotion of a common understanding of open science, scientific communication, national research and education networks and infrastructures for open science; Cuba do not have a plan, a strategy, a policy or a national roadmap for the effective adoption of practices for open science. The system policy on science, technology and innovation updated at the end of 2024, will serve as the framework for the development of the national science, technology, and innovation law that will improve the implementation of the Recommendation on Open Science.